Do Now · retrieval

Lesson 105 · RAM, ROM and Virtual Storage: Different Roles

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Where are running program instructions and data commonly held?

Q2

What does ROM or firmware flash retain?

Q3

What is a remote network drive an example of?

Q4

Rising edge D-type flip-flop D0 Q1: Q just after edge?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — RAM holds currently used programs/data and normally loses content without power; ROM or firmware flash retains startup routines.

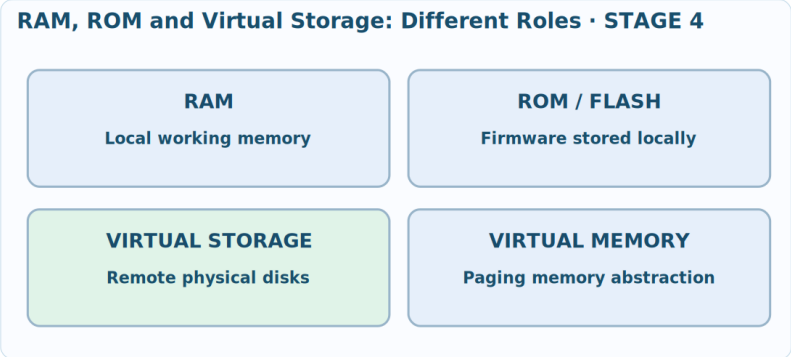
Core idea 2

Apply — OCR virtual storage means logical storage presented from physically remote or pooled devices, such as a network drive or cloud service.

Core idea 3

Evaluate — Do not confuse virtual storage with virtual memory. Virtual memory uses secondary storage to extend the apparent addressable memory of processes; virtual storage is remote-file presentation.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

RAM — Volatile working memory.

ROM — Non-volatile storage for relatively fixed code.

virtual storage — Remote/pooled physical storage presented logically.

virtual memory — OS memory abstraction using paging.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

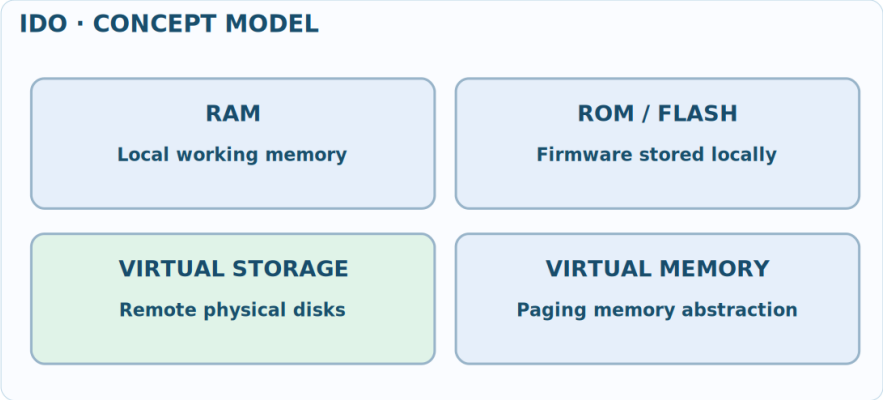
Problem

A pupil opens a document from a mounted school network drive. Where is the file physically stored?

Follow the worked method

1. Explain the choice — On one or more remote network storage devices, even if Finder/Explorer shows a drive like local storage.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the choice

No. It is virtual memory under 1.2.1, using secondary storage for memory pages.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain the role and volatility of RAM and ROM/firmware storage. [6]

Question 2

Define virtual storage and give an example, two benefits and a disadvantage. [8]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Correct a pupil's claim that 'virtual storage is extra RAM created when memory fills up'. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

OCR H446 1.1.3(d) virtual storage is another term for swapping RAM pages.

Use key vocabulary from memory

1. Explain RAM in your own words.

2. Explain ROM in your own words.

3. Explain virtual storage in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.